Chapter 9 exercises

Use $\langle u, v\rangle = u^H v$, with conjugation in the first argument. If choosing the opposite convention, rewrite every inner-product formula consistently.

CH09-E1

1. Compute $(2 - i)(3 + 2i)$ and $(2 - i)/(3 + 2i)$.
2. Express $1 + i\sqrt{3}$ in polar form.
3. Compute $(1 + i\sqrt{3})^6$.
4. Find all four roots of $z^4 = -16$ and locate their arguments.

CH09-E2

Let

$$u = (1 + i, 2 - i)^T, \qquad v = (2 - i, 1 + i)^T.$$

1. Compute $u^H u, v^H v, u^H v, v^H u$.
2. Verify $u^H v = \overline{v^H u}$.
3. Decide orthogonality under the stated convention.
4. Normalize u and check its norm.

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CH09-E3

Let

$$S = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix}.$$

1. Check that S is Hermitian and identify its real diagonal.
2. Find the characteristic polynomial and eigenvalues.
3. Find orthonormal eigenvectors.
4. Give $S = Q\Lambda Q^H$ and verify $Q^H Q = I$.

CH09-E4

For square $A \in \mathbb{C}^{n\times n}$,

1. Prove $(A^H)^H = A$ and $(AB)^H = B^H A^H$ for compatible square matrices.
2. If $A^H = -A$, prove its eigenvalues have zero real part.
3. If A is unitary, prove $|\det A| = 1$.
4. Determine the eigenvalue signs for a positive-definite Hermitian matrix.

CH09-E5

Let

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

1. Verify $Q^H Q = I$.
2. Check preservation of the inner product of $u = (1, 0)^T$ and $v = (0, 1)^T$.
3. Find the eigenvalues and check their moduli.
4. Determine whether Q is Hermitian.

CH09-E6

For $n = 4, \omega = e^{2\pi i/4} = i$.

1. Write F_4.
2. Compute $y = F_4 c$ for $c = (1, 2, 0, -1)^T$.
3. Verify inversion using $F_4^{-1} = F_4^H / 4$.
4. Check $\|y\|^2 = 4\|c\|^2$.

CH09-E7

1. For $n = 8$, show the even-odd split and both size-4 subproblems.
2. Write the butterfly and its twiddle index.
3. Derive $T(n) = 2T(n/2) + \Theta(n)$ and prove $T(n) = \Theta(n \log n)$.
4. Explain why radix-2 alone does not handle arbitrary lengths and give an alternative.

CH09-E8

1. Prove that both the rows and the columns of a unitary matrix form orthonormal bases.
2. Prove a Hermitian matrix has real trace and determinant.
3. Find all scalars c making cF_n unitary.
4. Compare the conventions $e^{2\pi i/n}$ and $e^{-2\pi i/n}$. Explain the forward/inverse labels and why the corresponding Parseval identity still holds.

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CH09-E9

Let $n = 4$, $c = (1, 2, 0, -1)^T$.

1. Give the bit-reversal permutation and reordered sequence.
2. Compute the two-point butterflies stage by stage, listing every intermediate value.
3. Use $\omega_4^0 = 1, \omega_4^1 = i$ to find $F_4 c$.
4. Directly evaluate y_1, y_3 to check the FFT.

CH09-E10

Use $\langle x, y \rangle = x^H y$, and let $A = \begin{bmatrix} i & 1 \\ 0 & 2 \end{bmatrix}$, $x = (1, 0)^T$, $y = (1, i)^T$. Compute A^H, Ax, and $A^H y$, and check $\langle Ax, y \rangle = \langle x, A^H y \rangle$ term by term. What happens if A^T is used instead?

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 9.

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